

Kidney Disease Prediction using ML techniques

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Abstract—Over the last two years there has been a wide progress in detecting causes of diabetic nephropathy, understanding its complications and preventing it. So far, techniques such as SVM, Decision Tree, KNN and ANN trained predictive models were used by researchers to monitor the progression of DN using T1- and T2-Weighted Imaging. We have discussed the importance of various Machine learning, Deep learning and artificial intelligence techniques used in curing diabetic nephropathy. Further a prospective of precise diagnosis of diabetic kidney disease is discussed to throw light on integrated bio-informatics analysis and machine learning algorithms. We also compare and analyze the research findings and solutions along with their challenges for a summarized view of the topic. In this research work, we have discussed about the Kidney diseases and the ML techniques wherever applicable.

Index Terms—Machine Learning algorithms, Diabetic Nephropathy, Bio-informatics.

I. INTRODUCTION

A rapidly growing kidney disease, known as Chronic Kidney Disease, is affecting millions of people due to untimely and unaffordable treatment. In 2013, near about a million people died due to this Chronic Kidney Disease [1]. The occurrence of this disease is due to the failure of both the kidneys and the patients suffer for longer period of time. Failure of kidney is due to improper functionality of the kidney and this improper functionality is due to any kind of disorder or due to lacking of Glomerular Filtration Rate [1].

One of the serious Chronic Disease of Kidney is Diabetes Mellitus. Due to this Diabetic Disease of kidney, the risk of mortality of diabetic patients increased and among them, majority were diagnosed by Type 2 Diabetes Mellitus.

Advanced techniques in microscopy and digital imaging are crucial in the identification and assessment of diverse kidney conditions [2]. The AI algorithms are used for the purpose of guiding Clinical practice, such as medical image analysis, prognosis prediction, smart diagnosis and curative effect evaluation.

Recently, machine learning models have proven their efficacy in various medical applications, demonstrating their value in analyzing, assessing, and predicting various medical conditions. In this research work, we reviewed the ML-based approaches to predict the presence of cardiovascular diseases in patients using essential health data.

II. LITERATURE REVIEW

The paper [3] developed an intelligent system known as clinical Decision Support System to predict the Diabetic Nephropathy at the early stage. The authors used feature selection method using GA to predict the features affecting Diabetic Nephropathy and trained predictive models using these selected features, and ML algorithms. The results of this research study indicated that age, HbA1c levels, DAP, SAP, FGR, and the duration of diabetes involvement were the most significant factors in forecasting the occurrence of Diabetic Nephropathy. In order to predict the DN, the highest performance was obtained by combining GA with the DT ID3 algorithm in terms of accuracy, sensitivity, specificity, and AUC, equals 98.9%, 98.6%, 99.2%, and 98.9%, respectively.

The author in this paper [4] proposed a kernel KFCM model along with two other algorithms – BCFCM (FCM algorithm which is based on neighbourhood pixel information), and EnFCM for comparison, for analysing the effects of image segmentation on these algorithms. The effects of correlation between renal cortex R2* termed as RC-R2*, renal medulla R2 as RM-R2*, renal cortex D as RC-D, renal medulla D as RM-D and renal function on the prognosis were compared. The results indicated the correlation coefficients between RC-R2* of 0.57, RM-R2* of 0.62, RC-D of 0.49, and between RM-D and renal function of 0.38.

The research work in paper [5] evaluated some machine models such as LR, extra tree classifier, RF, GBDT, XGBoost, and light gradient boosting machine. Based on these evaluation, the author concluded that the highest AUC, of 0.953

was achieved by XGBoost algorithm as compared to that of the extra tree acquiring 0.952 and GBDT acquiring 0.938 AUC values. In the XGBoost model, the SHapley Additive explanation summary plot acquired the five most important features that includes mean serum creatine, which is obtained within a timeline of one year before the diagnosis of T2DM, baseline serum creatinine, high sensitive C-reactive protein, female gender and protein-to-creatinine ratio of spot urine.

The researcher in this paper [6] used deep learning to develop a predictive score for kidney biopsy specimens and evaluated its performance against established standards and clinical decision support systems. The specifics of the architecture and training process involves selecting a suitable DL architecture (e.g., CNN, RNN), data preprocessing (e.g., resizing, normalization), and training with appropriate loss functions and optimization algorithms. The study enrolled a total of 442 patients. The paper concludes that among the MEST-C components such as mesangial, endocapillary, segmental sclerosis, and crescent parameters, only the endocapillary parameter had a significant impact on predicting kidney survival.

In order to assess the renal function in Diabetic Nephropathy, the authors [7] tried to find the possible practical scenario of the ML based mMRI-TA. The authors included 70 patients and randomly assigned 49 patient cases to training cohort and 21 to testing. They used the mMRI-TA model for classifying the groups and achieved robust and excellent performance with AUC scores of 0.978, 0.852 and 0.972, in the training cohort and 0.961, 0.809, and 0.850, respectively, in the testing cohort.

The authors in this paper [8] developed SRS risk score by using LASSO OPERATOR. The score is based on 5 hub genes and its mRNA levels are verified by RT-PCR. The researcher validated the SRS risk score and kidney function relationship by associating them with mitochondrial function and immune filtration.

A WGCNA was performed in this paper [9]. The researchers found out the key for DN progression as red and turquoise co-expression modules. They have utilized some ML algorithms such as random forest algorithm, SVM, AdaBoost, and KNN. Based on their analysis, the Random Forest model exhibited the highest AUC curve. They also observed some changes in immune infiltration patterns between Diabetic Nephropathy patients and control subjects.

The study [10] trained a CNN and fully designed Network to design a method for capturing the metabolic complexity and diagnose Diabetic Kidney Disease. The researchers used this method to diagnose the early stages of Diabetic Kidney Disease with accuracy as 83.3%, sensitivity as 83.2% and specificity as 80.2% along with the advanced stages of Diabetic Kidney Disease having accuracy 83.3%, sensitivity 82.9% and specificity as 81.0%. The researchers also used this method for diagnosis of Diabetes Mellitus with accuracy 83.3%, sensitivity 83.4% and specificity 80.4% and that of Chronic Kidney Disease having accuracy as 87.0%, sensitivity as 83.7% and specificity as 81.2%.

Leveraging the GEO database and the Limma R package,

the authors [11] pinpointed genes with differential expression in DN and obtained oxidative stress-related ones. Subsequently, we used a combination of WGCNA, ML techniques, and PPI networks to identify immune and oxidative stress-related hub genes. The correlation between hub genes and clinical characteristics was examined. To gain insights into the cellular distribution of the hub genes in DN, the researchers carried out single nucleus RNA sequencing. Through the combination of three ML algorithms through WGCNA analysis, this study unveiled three hub genes, specifically CD36, ITGB2, and SLC1A3, which holds promise as the main targets for the diagnosis.

The authors [12] employed DL techniques to identify a type of ARHGEF18 that is associated with a greater risk of both DKD and ESKD. This study seeks to leverage the integration of EHR data to uncover subphenotypes. Subsequently, they performed GWAS separately for different patients within each cluster. To further investigate the mechanistic implications of the novel ARHGEF18 variant, they conducted various experiments, including cytoplasmic, focal adhesion, and cytoskeletal morphometrics, live-cell motility assessments, Rho GTPase activity assays, and protein degradation experiments. Notably, Cluster S exhibited a higher prevalence of proteinuria compared to M cluster.

This paper [13] detected potential diagnostic markers for diabetic nephropathy and investigated the role of immune cell infiltration in this condition. The researchers utilized three gene expression datasets (GSE30528, GSE96804, GSE1009). They identified DEGs and then performed enrichment analyses on these DEGs. The findings revealed the strong associativity of DEGs with immune and the inflammatory responses and had close connections with the diseases related to kidney, urinary disease, and kidney cancer.

In this study, [14] the authors utilized DKD datasets obtained from the GEO database. They employed various analytical methods to investigate diabetic kidney disease (DKD) and identify diagnostic markers. The methods included ORA to understand the fundamental bio-processes for screening diagnostic markers of DKD. The assessment of the diagnostic model's reliability and applicability was conducted via curves such as calibration curves, ROC, and decision curve analysis (DCA). The study revealed that biological processes related to activation such as immune, T-cell activation, and cell adhesion were enriched in the DKD. The author identified some potential markers regarding diagnosis such as tenascin C, peroxidasin, tissue inhibitor of metalloproteinases 1, and tropomyosin for the DKD through various bioinformatics analyses.

The key steps used in this study [1] include feature selection using Pearson correlation, selecting the best machine learning models based on critical literature analysis, and combining these models into a hybrid framework. Specifically, Gaussian classifier of Naïve Bayes Algorithm, gradient boosting, and decision tree ID3 classifiers are included as base classifiers, while random forest is served as a meta-classifier in hybrid model. In the implementation phase, Gradient boosting

achieved approximately 99% accuracy, random forest achieved an accuracy of 98%, the decision tree classifier achieved 96%, and the proposed hybrid model outperformed others with accuracy of 100% on the same dataset.

The authors in this paper [10] utilized a dataset of 1521 clinical participants obtained through gas chromatography-mass spectrometry (GC-MS) to train a CNN and FC. They translated data to their responding images and applied normalization technique, enabling the DNN for extracting the features related to these diseases. The metabolite-based deep neural network achieved high accuracy, sensitivity, and specificity for diagnosing both early and advanced stages of DKD. It also demonstrated effectiveness in diagnosing DM and chronic kidney disease (CKD). Specifically, the accuracy for DKD diagnosis was 83.3% (early stage) and 83.3% (advanced stage), with corresponding sensitivities of 83.2% and 82.9% and specificities of 80.2% and 81.0%. Furthermore, it exhibited an accuracy of 83.3% for DM diagnosis and 87.0% for CKD diagnosis.

In this study, the researchers [8] obtained information about cellular senescence-related pathways and assessed their activity in DKD patients using GSEA algorithm. Furthermore, the study identified those module genes that are associated with cellular senescence pathways using the WGCNA algorithm. They constructed a cellular SRS risk score based on these hub genes using two methods such as the Least Absolute Shrinkage method and the Selection Operator method.

The authors [8] downloaded three datasets (GSE1009, GSE30528, and GSE47183) containing samples from both DN patients and normal individuals. They identified differentially expressed genes by comparing DN cases and control cases. To screen for potential diagnostic markers, the study utilized SVM-RFE analysis and the LASSO. Through machine learning approaches, the study identified C1orf21, TGFBI, ATF3, and BHMT as those genes for diagnosis of DN.

This study [15] used only non-targeted metabolomics that included plasma and the urine samples from those patients particularly having DKD. The research detected rapid decliners of kidney function by analyzing changes in eGFR over a 30-month period. The deep learning approach narrowed down 3388 variables to 50 and employed two regression models: piecewise linear model and handcrafted linear model. The DNN identified potential biomarkers, such as systolic blood pressure, urinary albumin-to-creatinine ratio, six known metabolites, and three unidentified metabolites, with urinary NMP among them.

The study [16] developed and validated a ML algorithm capable of predicting the severity of the stages of DKD that ranges within a timeline of 5 years of a T2DM diagnosis. Two ML algorithms were trained to predict the severity of DKD at different stages and were compared with the CDC risk score. The results indicated that the ML algorithms outperformed the CDC risk score both in the hold-out test and externally collected datasets. Specifically, the algorithms scored an AUROC of 0.75 for predicting any stage of DKD and a score exceeding 0.82 of AUROC for more severe DKD

endpoints. In contrast, the CDC risk score had an AUROC below 0.70.

We have systematically represented the above review in a tabular form for better understanding and easy interpretation in Table I.

Year	Machine Learning Algorithms Used	Accuracy
2022	SVM, Decision Tree ID3, KNN and ANN [3]	Combining GA with DT ID3 - 98.9% accuracy, 98.6% sensitivity, 99.2% specificity, and 98.9% AUC
2023	logistic regression classifier, extra tree classifier, RF algorithm, GBDT, XGBoost, and light gradient boosting [5]	0.953 AUC achieved by the XGBoost algorithm and that of extra tree and GBDT acquiring 0.952 and 0.938 values.
2023	LASSO [8]	NA
2023	RF classifier, SVM classifier, Adaptive boosting algorithm (AdaBoost), and KNN [9]	NA
2023	Convolutional Neural Network [10]	early stages of Diabetic Kidney Disease with 83.3% accuracy, 83.2% sensitivity and 80.2% specificity along with the advanced stages of Diabetic Kidney Disease having 83.3% accuracy, 82.9% sensitivity and 81.0% specificity.

TABLE I
TABULAR ANALYSIS

III. METHODOLOGY

In order to predict and analyze the disease caused under Nephropathy section, we are working on Diabetic Nephropathy dataset so as to predict the type of kidney disease depending upon the factors. While doing the analysis work, we collected the datasets from the Kaggle Data Repository. We will perform some Machine Learning Algorithms to check the accuracies of each of the Machine Learning model.

A. Description of the Algorithms Used:

Support Vector Machine – A powerful supervised learning used for both classification as well as regression tasks. Vladimir Vapnik and his team in 1995 developed SVM [17]. In its simplest form, SVM works by finding the optimal hyperplane that best separates classes in a linearly separable dataset. This hyperplane serves as a decision boundary between different classes. However, the data, if not linearly separable, SVM uses what's known as kernel functions, to map the data, from the current dimensional space into a higher-dimensional space where it might become linearly separable. SVMs are particularly known for their ability to find the optimal separating hyperplane and their robustness against overfitting in high-dimensional spaces, making them a popular choice in machine learning.

Logistic Regression – A machine learning technique widely employed for both classification as well as regression tasks. It is capable of predicting binary outcomes (0/1) by leveraging a set of continuous and discrete predictors [18],[19]. LR finds extensive application in predicting various conditions such as diabetes, identifying spam emails, detecting cancer, and diagnosing other diseases.

Artificial Neural Network – Artificial Neural Networks (ANNs) represent a machine learning methodology applicable to both classification and regression tasks [20]. ANNs are analogous to the human brain, built upon neurons as their fundamental components. These networks consist of three layers such as the input layer, the intermediate or the hidden layers, and lastly the output layer.

K Nearest Neighbor – The KNN algorithm was initially developed by Fix and Hodges in 1951. It operates as a simple technique to classify new objects by assessing their neighbors through a distance function and determining the majority vote among them [21].

Decision Tree – A supervised machine learning tool applied for both classification as well as regression tasks. It takes the form of a flowchart-like structure where every internal node signifies a feature or an attribute, each branch describes a decision rule, and every leaf node signifies the outcome or the target variable [22]. The primary objective of the ID3 classifier is to partition the dataset into subsets by focusing on the most impactful features that effectively distinguish between various classes or forecast the target variable. This recursive process persists until the algorithm determines the optimal approach to categorize the data or predict the target variable within each subset.

B. PreProcessing

Firstly, we have converted the Alphabetical Class types to Numeric data types. For Example, Yes converted to 1 and NO converted to 0. The conversion is done to convert the dataset into numerical dataset.

Then by using the heatmap() function, we have checked the missing data of the dataset as shown in in Fig No.1. Finally we have cleaned the dataset by dropping all the records having NAN values or missing values.

IV. EXPERIMENTAL RESULTS

We then performed some ML Classification Algorithms including DT Algorithm, SVM classifier, KNN Classifier, Logistic Regression. After performing these classification Algorithms, we have found out the accuracies of these algorithms to verify which algorithm suits best for the given dataset. The accuracies of the algorithms are mentioned in Table II.

V. CONCLUSION AND FUTURE WORK

Diabetic Nephropathy is a severe complication arising from diabetes, characterized by gradual kidney structure damage, resulting in declining kidney function and an increased risk of final stage of the kidney disease or permanent damage of the kidney. This condition develops progressively over time,

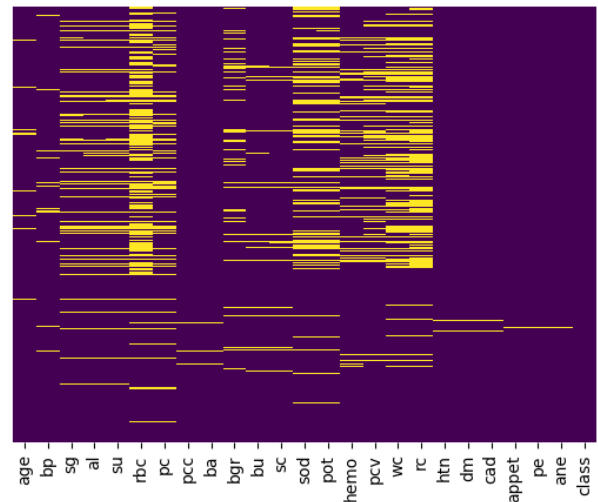


Fig. 1. Heatmap showing missing values

Algorithms Used	Accuracies	Dataset Used
Artificial Neural Network	71%	kidney_disease
Decision Tree	96.88%	kidney_disease
KNN Classification	83.33%	kidney_disease
Logistic Regression	96.88%	kidney_disease
SVM Implementation	97%	kidney_disease

TABLE II
EXPERIMENTAL ANALYSIS

and when identified in its early stages, it can be effectively monitored and treated. Moreover, the ability to predict the potential development of this disease offers a valuable opportunity for interventions to delay its onset. This review primarily focuses on approaches that utilize clinical data. Based on the review, it has been noted that an accuracy of 71% has been achieved for ANN. It is worth noting that the studies in this field vary in terms of population, data types and volume, outcomes, and research goals, which can introduce limitations in the findings of this review. Future research should aim to address these limitations by introducing image datasets thereby implementing Deep Learning Algorithms to help improve Kidney Disease or Kidney Failure prediction more accurately in coming years.

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